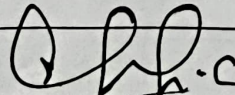


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PROJECT CENTRIC LEARNING (PCL) PHASE 2

			PROJECT TEAM NO.	CoEPCL-0216
PROJECT TITLE		SMART WASTE MANAGEMENT (ECOSTREAM- AI)		
TEAM MEMBERS				
S. No.	USN No.	NAME OF THE STUDENT	CLASS	MOBILE NO.
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DETAILS OF THE PROJECT GUIDE				
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DOMAIN OF THE PROJECT				
Artificial Intelligence / Computer Vision / Sustainable IoT				
Scope of the Project				
1. Material Recovery Enhancement: Providing 94% accurate real-time classification to reduce contamination at the source.				
2. Regulatory Compliance: Scaling the RAG corpus to include every major Indian municipality's specific solid waste bylaws.				
3. Cost Reduction: Enabling a 35% reduction in collection costs by replacing hardware sensors with software predictive models.				
4. Behavioral Change: Incentivizing proper disposal through an interactive AI that educates users in real-time rather than just monitoring them.				


PROJECT GUIDE**PROJECT COORDINATOR**

Existing System:-

The existing waste management landscape in urban India is characterized by a fundamental reliance on manual labor and rigid, inefficient logistics. Despite the "Smart Cities" initiative, the prevailing systems continue to struggle with a significant technological and operational "Reality Gap".

1. The High Cost of Physical Infrastructure

Current "Smart Bin" solutions primarily utilize ultrasonic sensors to monitor fill levels, a strategy that is financially unviable for large-scale urban deployment. Implementation costs for these IoT-based systems average approximately \$120,000 per 1,000 bins. For a tier-1 Indian city requiring tens of thousands of bins, this massive capital expenditure (CapEx) creates a prohibitive barrier to entry.

2. Environmental Fragility and Vandalism

The physical environment of Indian streets is notoriously harsh for electronic components. Traditional sensors frequently succumb to moisture during heavy monsoon seasons and chemical corrosion from waste leachate. Furthermore, public infrastructure is highly vulnerable to vandalism and theft, leading to high maintenance costs and frequent system downtime.

3. The Logistical "Reality Gap"

There is a stark discrepancy between simulated routing efficiency and real-world performance. While simulated models report distance reductions of nearly 40%, real-world implementations in chaotic urban environments achieve only about 12%. This failure stems from a reliance on static data points that ignore dynamic variables such as traffic congestion, road closures, and unpredictable waste generation spikes.

4. Passive Monitoring and Segregation Failure

Perhaps the most critical failure is the passive nature of current monitoring. Fill-level sensors cannot distinguish between recyclable plastic, hazardous e-waste, or organic matter. This lack of real-time material identification leads to high contamination rates at the source, making downstream recycling economically impossible. Furthermore, these systems provide no educational feedback to citizens, failing to address the human behavioral component of waste management.

Problem Identification

Segregation Failure: Lack of real-time material identification leads to high contamination in recycling streams, making recovery economically unviable.

Information Asymmetry: Citizens often do not know the specific municipal bylaws (e.g., SWM Rules 2016) for different materials in their specific city.

Operational Inefficiency: Municipalities lack predictive data, leading to "overflowing bins" in some areas and "empty runs" for collection trucks in others.

Hallucinations in AI: General-purpose AI models often provide incorrect or "hallucinated" advice regarding specific industrial codes or local regulations.

Proposed System:-

EcoStream AI transitions waste management from a reactive, hardware-centric model to a proactive, "Software-First" intelligent ecosystem. By leveraging existing mobile and vehicular camera infrastructure, the platform achieves zero-additional-hardware scalability while addressing the core failures of traditional IoT sensors.

Core Technical Pillars

- **Real-Time Edge Vision:** The system utilizes a YOLOv11s-seg instance segmentation model, which provides a significant 2.8 percentage point improvement in mAP over traditional detection baselines. Unlike standard bounding boxes, this model achieves pixel-level precision, allowing it to segment six distinct waste classes even when items are heavily soiled, crushed, or overlapping in cluttered urban environments.
- **Grounded AI Advisory (RAG):** To eliminate AI hallucinations, EcoStream AI implements a Hybrid RAG pipeline that combines dense BGE embeddings with BM25 sparse search and cross-encoder reranking. This ensures that disposal advice is strictly anchored in a verified regulatory corpus, including the National SWM Rules 2016 and specific municipal bylaws for Bangalore, Mumbai, Delhi, and Chennai.
- **Predictive Analytics:** The platform replaces fragile fill-level sensors with a predictive engine using Gaussian Kernel Density Estimation (KDE) and exponential time-decay forecasting. By analyzing historical 14-day scan patterns and applying day-of-week multipliers, it forecasts next-day waste hotspots, allowing municipalities to optimize collection routes before bins overflow.
- **Integrated Architecture:** Orchestrated via Docker Compose, the microservices stack (FastAPI, React, PostgreSQL) delivers an average end-to-end latency of just 979ms, making real-time field operations viable.

No. of Modules

Sl. No.	Name of the Module	Remarks
1	Vision Classification Engine	Employs YOLOv11s-seg to achieve real-time material identification with pixel-level instance segmentation.
2	RAG Policy Advisor	Anchors AI-generated disposal guidance in India's SWM Rules 2016 to ensure high factual accuracy.
3	Geo-Spatial Heatmap	Visualizes uncollected waste intensity using Gaussian KDE to identify immediate urban hotspots.
4	Citizen AI Chatbot	Facilitates multi-modal interaction for disposal advice via a hybrid dense-sparse retrieval pipeline.
5	Municipal Command Center	Provides an administrative interface for zone resolution and real-time material analytics.
6	Hotspot Predictor	Utilizes exponential time-decay and day-of-week multipliers to forecast next-day waste accumulation.


PROJECT GUIDE

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